



AINS HACKATHON 2026 · AI FOR ENTREPRENEURSHIP

Intelligent Entrepreneurial Orientation Engine

Project-maturity diagnosis · explainable multi-dimensional scoring · personalised roadmaps for entrepreneurs in Tunisia.

FIRST SUBMISSION · WORKING DRAFT

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The Problem

Entrepreneurs can't see where their project actually stands — so every orientation tool reinforces their own blind spots.



Misjudged maturity

Founders systematically over- or under-estimate their stage. A "financing-ready" founder often has no validated business model.



Fragmented information

Guidance exists but is scattered, generic, and never contextualised to a specific project profile.



Misaligned orientation

Entrepreneurs are sent to programs they don't qualify for and resources that don't match their real stage.

Existing tools answer questions — they do not diagnose, score, or produce an evidence-based assessment.

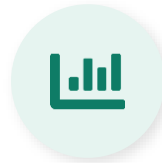
Our Solution

A multi-agent AI platform that transforms fragmented project data into an explainable maturity diagnosis, multidimensional assessment, and personalised entrepreneurial roadmap.



1. DIAGNOSE

Maturity stage classification + perception-reality gap detection.
Adaptive intake that branches on prior answers.



2. SCORE

Five composite scores (Market, Commercial, Innovation, Scalability, Green), each with weighted, justified sub-criteria.



3. GUIDE

Personalised roadmap grounded in real Tunisian resources. Every recommendation traceable to a KB source.



Cross-module integration: A diagnostic gap triggers resource retrieval; a low sub-score surfaces targeted roadmap actions. The modules interact — they don't just coexist.

Who It's For

Primary context: entrepreneurship in Tunisia and the MENA region.



Primary Focus

Young Tunisian founders

Young Tunisian founders and unemployed graduates exploring entrepreneurship as a career path — need an objective assessment of project maturity, readiness, and the next actions required to grow.



Support-program officers

Triage applicants with evidence-based diagnosis instead of self-reported claims.



Ecosystem & investors

Consistent, explainable scoring to compare projects across dimensions.

System Architecture

TECH STACK

Frontend Next.js + Tailwind

Backend FastAPI

Orchestration LangGraph

LLM Qwen / Llama

Database PostgreSQL

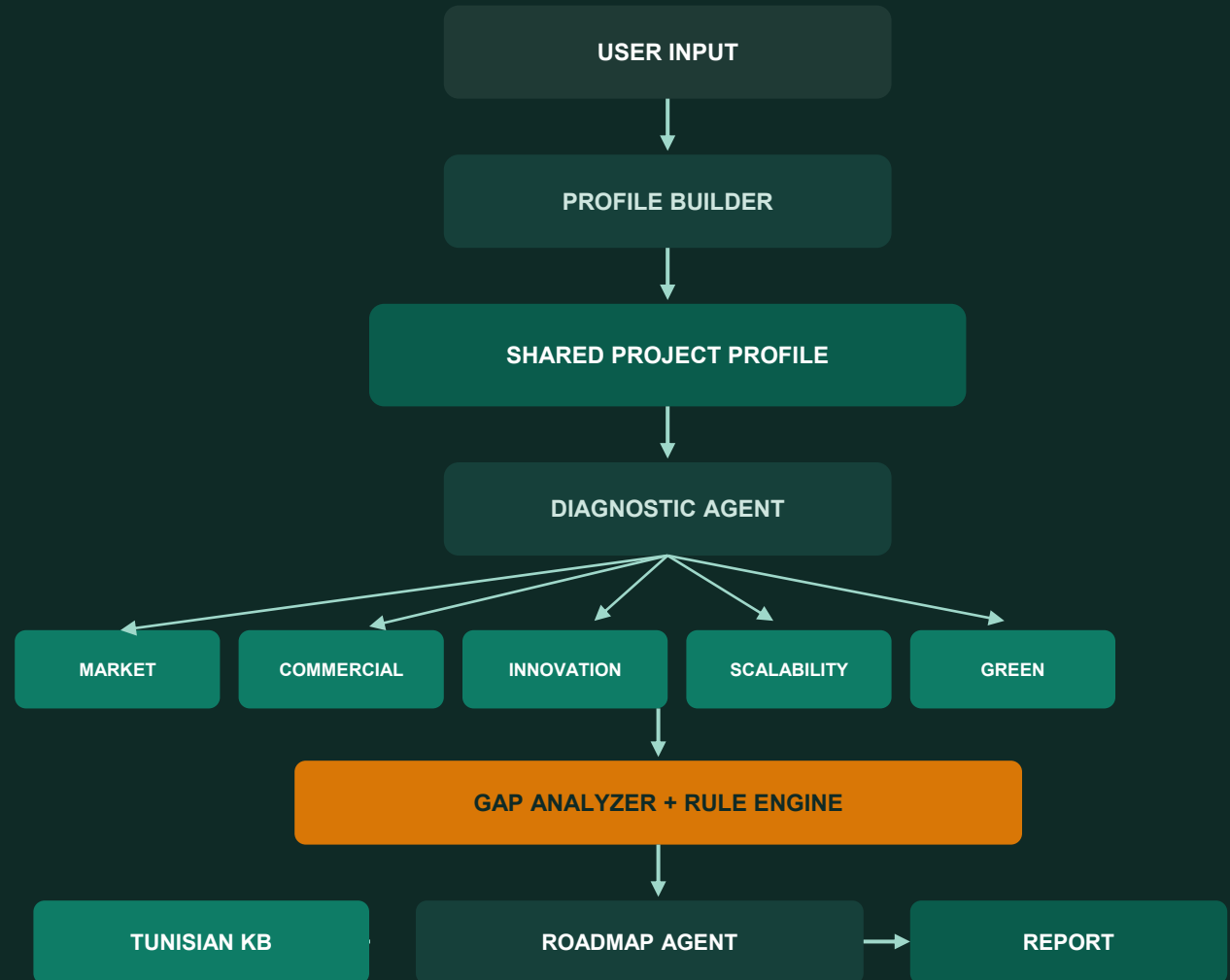
Vector DB Qdrant

RAG LangChain

Multi-Agent System

- ✓ Modular — each agent has a single responsibility
- ✓ Explainable — outputs traceable to rules
- ✓ Parallel scoring — reduces latency
- ✓ Maps directly to evaluation dimensions

System Pipeline



Multi-Agent System

Seven specialised agents on one shared profile:

1Diagnostic

2Market

3Commercial

4Innovation

5Scalability

6Green

7Roadmap

Taxonomy Criteria Model

BUILT

1Ideation

Default – no criteria

2Market Validation

has_validated_problem

3Structuration

business_model_documented · team_core_complete · legal_form_started

4Fundraising

has_paying_customers · financial_docs_exist · legal_form_registered

5Launch Planning

funding_or_self_financing · distribution_channel_tested

6Growth

recurring_revenue (3+ mo) · client_base_beyond_pilot

True

satisfied

False

real gap

None

missing data — surfaced, not hidden

Scoring Dimensions

IN PROGRESS

Five composite scores. Sub-criteria and weights are the modelling work — partially drafted below.



Market

Market share · customer validation · revenue model viability

weights: [__ / __ / __ / __]



Commercial Offer

Value-prop clarity · product readiness · pricing · offer-need fit

weights: [__ / __ / __ / __]



Innovation

Local novelty · tech intensity · barrier to entry · departure from existing offerings

weights: [__ / __ / __ / __]



Scalability

Replicability · manual dependency · deployment cost · addressable reach

weights: [__ / __ / __ / __]



Green

Environmental impact · SDG alignment · resource efficiency · circular economy

weights: [__ / __ / __ / __]

Aggregation logic

Composite scores are not simple averages. A weak score on a fundamental dimension must not be masked by strong scores elsewhere.



The Differentiator: Perception-Reality Gap

Detecting divergence between what a founder believes and what the diagnosis shows — designed for explicitly, not as an afterthought.

Self-assessment

“We're ready for financing.”

Founder believes the model is validated and the project is scalable.



System diagnosis

Actual stage: Structuration.

No validated business model; scalability depends on manual accompaniment. Gap flagged with the dimensions that caused it.

Target: correctly flag ≥ 3 divergence cases with a clear explanation of each gap.

Knowledge Base — Tunisian Resources

IN CONSTRUCTION

Real, source-traceable corpus for the RAG layer. Structured catalogue underway toward the 30+ target.

Covered so far (real institutions)

- ✓ **Support & accompagnement:** APII pépinières, ANPE, ODC, incubators
- ✓ **Financing:** BFPME, BTS, Startup Act funds, Flat6Labs
- ✓ **Administrative:** Startup Label, RNE, guichet unique, CEPEX
- ✓ **Ecosystem:** CONECT, IEEE, Wiki Startup, AINS 4.0

23

entries structured

Schema: name · provider · category · eligibility · stage · amount · how-to-apply · source URL



How it connects: The Roadmap Agent uses the project profile, maturity stage, and scores to retrieve relevant Tunisian resources and generate personalised next steps. Every recommendation cites its KB source.

Road to Final Submission

Current completion status and remaining milestones.

Delivered now

- ✓ Problem framing & target users
- ✓ 6-stage taxonomy criteria model (True/False/None)
- ✓ 5 scoring dimensions defined
- ✓ KB started — 23 real entries
- ✓ Architecture & integration plan

Next — to assign / complete

- ✚ 1. Finalize scoring weights
- ✚ 2. Build diagnostic classifier
- ✚ 3. Complete KB (30+ resources)
- ✚ 4. Implement RAG roadmap
- ✚ 5. Build dashboard
- ✚ 6. Evaluate on labeled dataset